

UNIVERSITY OF CAPE TOWN
DEPARTMENT OF STATISTICAL SCIENCES - STA2020F: APPLIED STATISTICS
EXAM – 9 June 2025

Time: 180
MINUTES

Max Marks: 100

Examiners: Ms. G. Carmichael
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**Internal
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**External
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Swanepoel

Instructions: Answer ALL questions.
Marks are awarded for intermediate calculations.
Answer all MCQ questions in **SECTION A** *IN PENCIL* on the MCQ sheet provided.
Answer all written questions *IN PEN* in the answer booklet(s) provided.
Answer each written question on a *NEW PAGE* in your answer book.
Answer **SECTION B** in the first booklet, **SECTION C** in the second booklet, and **SECTION D** in the third booklet.
ROUND ALL ANSWERS TO 2 DECIMAL PLACES UNLESS OTHERWISE SPECIFIED.
DO NOT ROUND INTERMEDIARY CALCULATIONS.

SECTION A: MCQ QUESTIONS

[20]

1. Fifty observations are recorded for the variables X and Y, and the following values are computed:

$$\sum(x_i - \bar{x})^2 = 8541$$

$$\sum(y_i - \bar{y})^2 = 7124$$

$$\sum(x_i - \bar{x})(y_i - \bar{y}) = 6528$$

What is the value of the Pearson correlation coefficient for X and Y? (2)

- A. 0.84
- B. 0.52
- C. 0.95
- D. 0
- E. We cannot calculate the Pearson correlation coefficient using this information.

2. A group of researchers collects data on the height and age of a group of infants. Some of these infants were born with a congenital disease, and the researchers are interested in studying the infants' height. Specifically, the researchers are interested in whether the relationship between an infant's height and age is different for infants with and without the disease. Which of the following regression models would you fit to answer the researchers' question? (2)

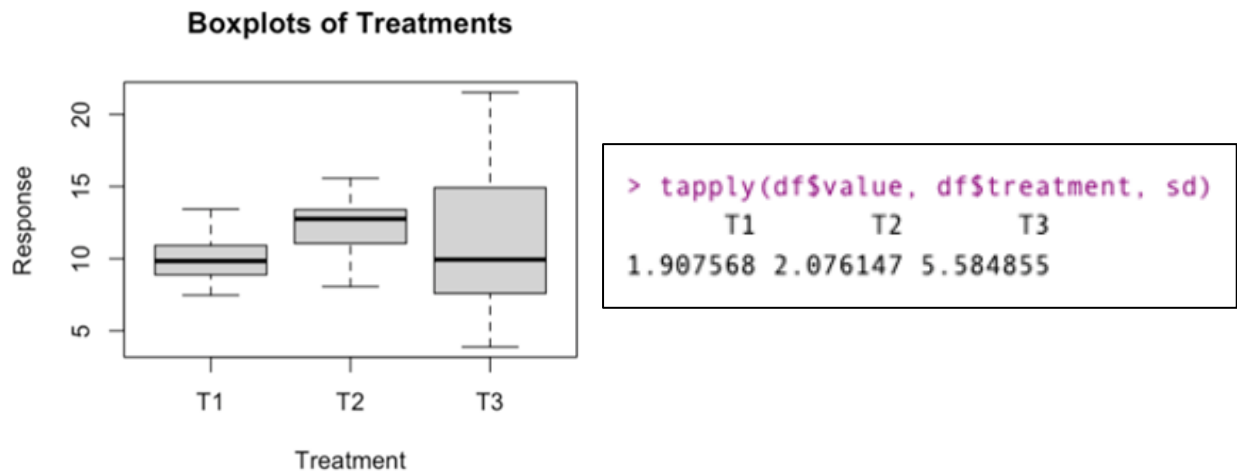
- A. $height_i = \beta_0 + \beta_1 \times age_i + \beta_2 \times disease_i + \varepsilon_i$
- B. $age_i = \beta_0 + \beta_1 \times height_i + \beta_2 \times disease_i + \varepsilon_i$
- C. $age_i = \beta_0 + \beta_1 \times height_i + \beta_2 \times disease_i + \beta_3 \times disease_i \times height_i + \varepsilon_i$
- D. $\log - odds(disease_i) = \beta_0 + \beta_1 \times age_i + \beta_2 \times height_i + \varepsilon_i$
- E. $height_i = \beta_0 + \beta_1 \times age_i + \beta_2 \times disease_i + \beta_3 \times disease_i \times age_i + \varepsilon_i$

3. The table below contains the AIC values for 5 different regression models. Which of the following statements is **FALSE**? (2)

	Formula	AIC
Model 1	$y \sim x_1$	450
Model 2	$y \sim x_1 + x_2$	411
Model 3	$y \sim x_3$	426
Model 4	$y_2 \sim x_1 + x_3$	401
Model 5	$y \sim x_2$	433

- A. Model 2 is a better model than Model 1.
 - B. The variable x_3 is a better predictor of y than x_1 .
 - C. Adding the variable x_2 to Model 1 improves the AIC.
 - D. Model 4 is the best of the five models.
 - E. We cannot use the AIC to compare all these models.
4. Which of the following statements best describes the concept of replication in the context of experimental design? (2)
- A. Repeating the entire experiment under identical conditions.
 - B. Assigning the same treatment to multiple experimental units to estimate variability.
 - C. Using more than one treatment factor in an experiment.
 - D. Measuring the same experimental unit multiple times.
 - E. Blocking units based on a nuisance variable to reduce variability.

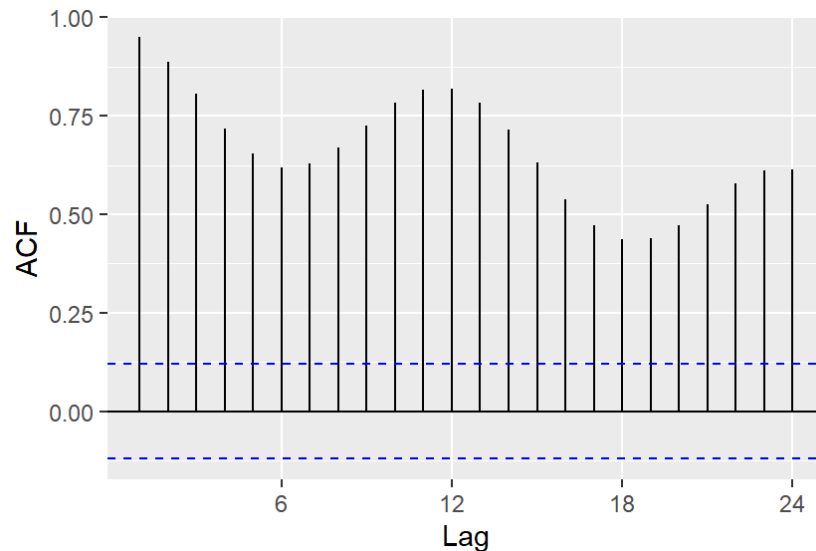
5. The box plots, together with the code below, are used to check an assumption of ANOVA. Which assumption is being assessed, and is it reasonable to conclude that the assumption holds? (2)



- A. The assumption of normality is being assessed. The symmetric shape of the boxplots suggests that this assumption is reasonable.
- B. The assumption of equal population variances is being assessed. The much larger spread and standard deviation in T3 suggest this assumption is likely violated.
- C. The assumption of equal population variances is being assessed. The code and output confirm that this assumption is reasonable.
- D. The assumption of normality is being assessed. The R output suggests that the spread of data is comparable between groups.
- E. None of the above statements is correct.
6. In a two-way ANOVA without replication (i.e., one observation per treatment), which of the following statements is TRUE? (2)
- A. Interaction can be estimated and tested separately from the main effects.
- B. The model is overparameterised and cannot be fitted.
- C. The interaction effect is confounded with the error term.
- D. The F-test for main effects is based only on total variation.
- E. Residuals cannot be calculated for any model with two factors.
7. When deciding between a parametric test (e.g., a t-test) and a non-parametric alternative (e.g., the Mann-Whitney U test), which of the following statements is most accurate? (2)
- A. Parametric tests should be avoided even if assumptions are only mildly violated, as they become less powerful.
- B. Non-parametric tests are preferred when the goal is to detect smaller differences between group medians.
- C. Parametric tests tend to be more powerful than non-parametric tests when assumptions like normality and homogeneity of variance are satisfied.

- D. Non-parametric tests are more powerful because they do not rely on sample statistics like the mean and standard deviation.
- E. Both test types perform equally well when sample sizes are large due to the central limit theorem.

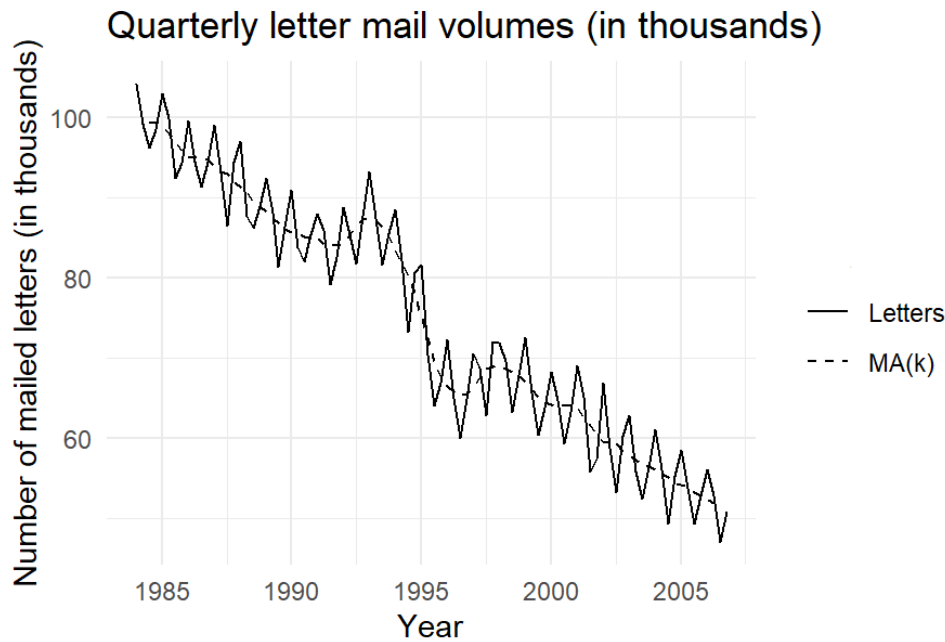
8. The following correlogram is plotted for a time series on the number of monthly subscriptions to a certain magazine in South Africa. Investigate the correlogram, consider statements (i) to (vi), and choose the correct option between statements A to E. (2)



- i) The correlogram indicates that we could expect the time series on the number of magazine subscriptions to have seasonality but no trend.
- ii) The correlogram indicates that we could expect to see a trend, as well as seasonality, in the time plot of the number of magazine subscriptions.
- iii) We expect the length of the seasonal variation to be one year, containing twelve seasons. The positive autocorrelation between y_2 and y_8 is approximately 0.625.
- iv) We expect the length of the seasonal variation to be one year, containing four seasons. The positive autocorrelation between y_6 and y_{17} is greater than 0.75.
- v) The correlogram indicates that the time series has a trend, but no seasonality.
- vi) The correlogram indicates that the time series has a trend and that it is linearly increasing.

- A. Only statements (ii) and (iii) are correct.
- B. Only statement (v) is correct.
- C. Only statements (ii) and (iv) are correct.
- D. Only statements (v) and (vi) are correct.
- E. Only statements (i) and (iii) are correct.

9. The following time plot contains observations on quarterly letter mail volumes. Investigate the time plot, consider statements (i) to (v), and choose the correct option between statements A to E. (2)



- i) The time series exhibits a decreasing linear trend.
- ii) There is seasonality, and the length of the seasonal variation is three months.
- iii) The time series exhibits cyclical variation around the years 1991 to 1997.
- iv) The time series contains no indication of cyclical variation. We can smooth out the seasonality by applying CMA(4) smoothing to the time series.
- v) The time series exhibits a decreasing nonlinear trend.

- A. Only (ii), (iii), and (iv) are correct.
- B. Only (iv) and (v) are correct.
- C. Only (i) and (iv) are correct.
- D. Only (i) and (iii) are correct.
- E. Only (iii), (iv), and (v) are correct.

10. The following output was generated from classical decomposition in R and represents the adjusted seasonal indices of the time series on quarterly letter mail volumes. Investigate the output. Calculate and interpret for $A = \hat{S}_3$. (2)

decomp_model\$seasonal			
Qtr1	Qtr2	Qtr3	Qtr4
1.066568	1.000392	A	1.000890

- A. On average, letter mail volumes in the third quarter are 6.79% above the annual average number of letters mailed.
- B. On average, letter mail volumes in the third quarter are 2.26% above the annual average number of letters mailed.
- C. On average, letter mail volumes in the third quarter are 2.26% below the annual average number of letters mailed.
- D. On average, letter mail volumes in the third quarter are 6.79% below the annual average number of letters mailed.
- E. On average, letter mail volumes in the third quarter are 93.22% below the annual average number of letters mailed.

SECTION B: TIME SERIES

[23]

QUESTION 1

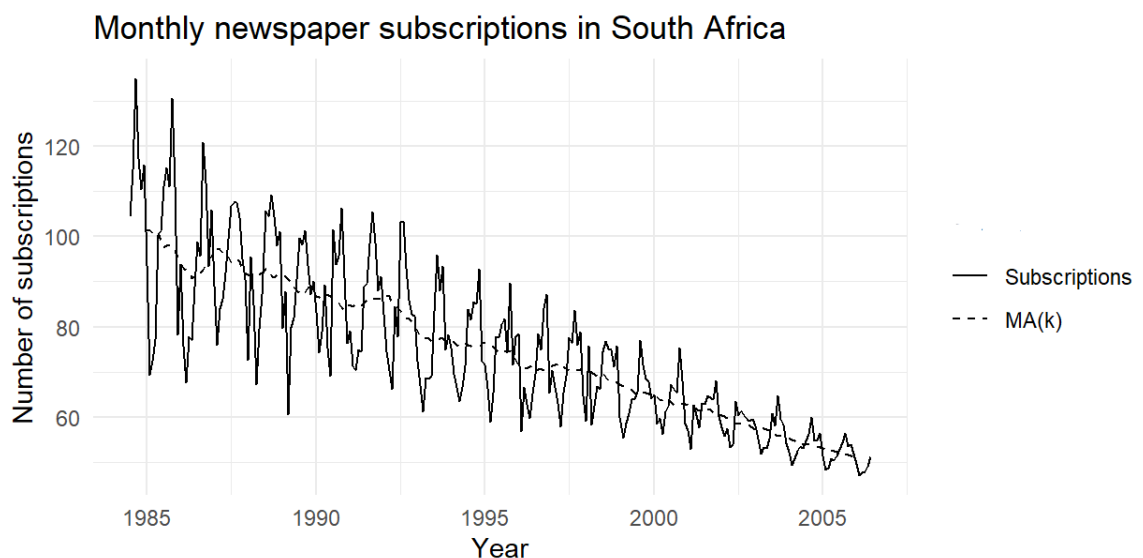
[16]

Data analysts at News24 have determined that over the past few decades, there has been a major shift in how people gain access to national and international news. One team member was tasked with investigating a time series on monthly newspaper subscriptions in South Africa for the purpose of forecasting how the subscription numbers would most likely look within the next 5 years.

- a) Calculate values **B** to **F** from the given snippet of time series data. (5)

Year	Month	t	Y_t	$MA(k)_t$	$CMA(k)_t$	De-trended data	Adjusted seasonal index	Seasonally adjusted time series
1985	2	8	69	---	---	---	0.89	101.57
	3	9	72	---	---	---	0.86	100.59
	4	10	77	---	---	---	0.88	100.14
	5	11	100	---	100.72	---	0.93	100.73
	6	12	101	97.42	99.19	---	0.98	99.19
	7	13	111	97.33	C	---	1.08	---
	8	14	115	B	---	---	1.11	---
	9	15	111	97.50	97.97	---	1.13	F
	10	16	130	---	97.80	---	1.13	97.80
	11	17	111	---	96.87	---	1.06	96.87
	12	18	78	---	95.38	D	1.00	95.38
1986	1	19	93	---	94.35	---	0.94	94.35
	2	20	75	---	93.03	---	E	93.03

- b) Investigate the time plot of the monthly number of newspaper subscriptions.



- i) Explain in as much detail as possible why we would not consider this time series to be stationary. (2)
 - ii) Would you expect a stronger autocorrelation at lag $g = 6$ or at lag $g = 12$? Explain your reasoning. (1)
 - iii) Which long-term simple forecasting method is bound to deliver the most accurate forecasted values? Explain your reasoning. (1)
- c) Suppose the data is split into a training (80%) and test (20%) set. Two models, MOD1 (the most accurate simple forecasting model for long-term forecasting) and MOD2 (the most accurate exponential smoothing model), were fit to this time series.

OUTPUT FOR MOD1:

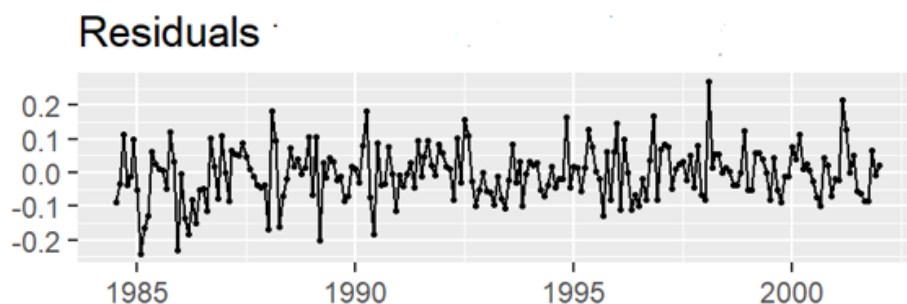
Forecasts:	
	Point Forecast
Feb 2002	57.37503
Mar 2002	57.15219
Apr 2002	56.92934

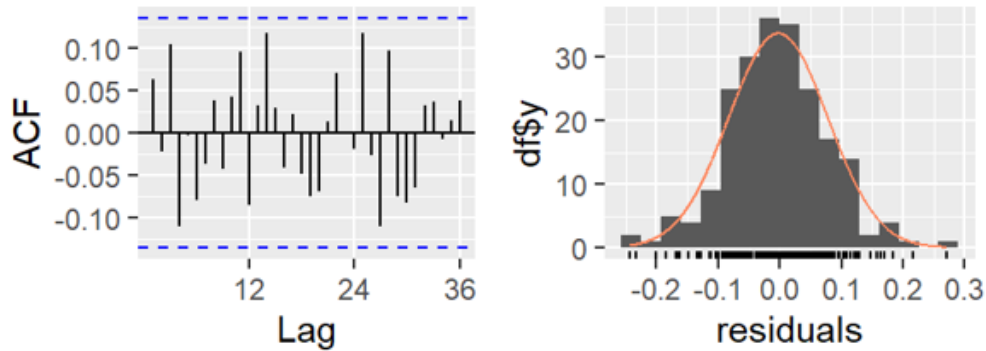
	ME	RMSE	MAE	MPE	MAPE	MASE
Training set	7.071741e-15	10.194796	7.724158	-0.7646498	9.610891	1.0079390
Test set	3.021452e+00	4.459223	3.668264	5.2947617	6.533416	0.4786782

OUTPUT FOR MOD2:

Forecasts:	
	Point Forecast
Feb 2002	52.44115
Mar 2002	49.89850
Apr 2002	51.45025

	ME	RMSE	MAE	MPE	MAPE	MASE
Training set	-0.3819921	6.568044	5.012411	-0.9585191	6.334883	0.6540783
Test set	0.5006325	3.780715	3.176971	1.1586337	5.833887	0.4145686





```

Ljung-Box test

data:  Residuals
Q* = 20.794, df = 24, p-value =
0.6508

Model df: 0.    Total lags used: 24

```

- i) Which model would you expect MOD2 to be? (1)
- ii) Using MOD1, compute an 80% prediction interval for the first forecasted value of the test set. The residual standard error is 0.7052. (1)
- iii) Which model between MOD1 and MOD2 is the preferred model to forecast future observations? Refer to the appropriate values in the given R output to explain your choice. (1)
- d) Investigate the residuals of MOD2. Give a detailed interpretation of the residuals. If necessary, make use of a 5% level of significance. (4)

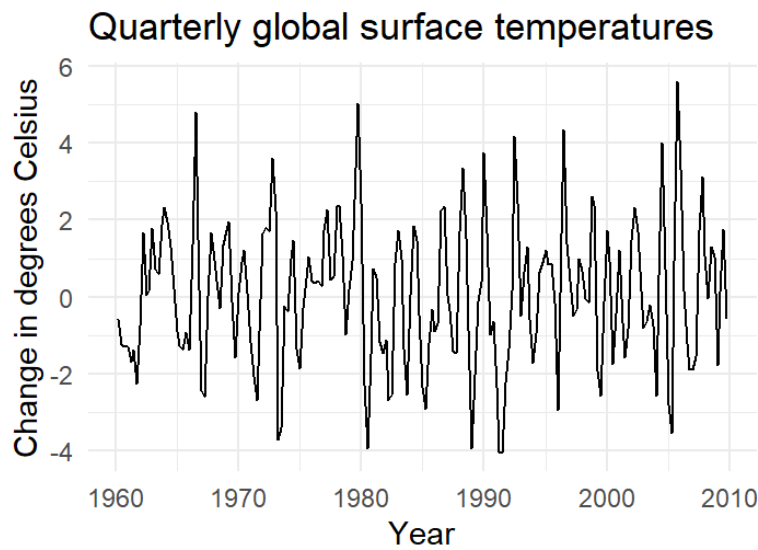
QUESTION 2

[7]

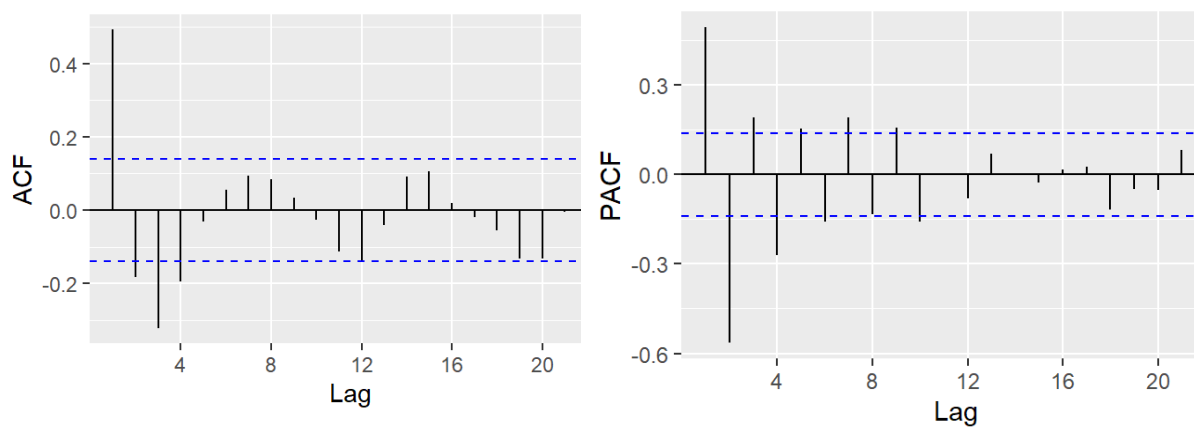
A company specialising in clean energy investigates time series on quarterly global surface temperatures, measured in degrees Celsius, between the years 1960 and 2010. They have noticed a steady rise in temperatures over time and need to make forecasts about what they think could potentially happen over the next 20 years. The team of in-house data analysts suggests that an ARIMA model would most likely be suited to this kind of data. You may assume that it was found that there was no clear indication of seasonality, and that a Box-Cox transformation revealed $\lambda = 1.0006$.

- a) Fully explain the $AR(p)$ part of an $ARIMA(p,d,q)$ model. (1)
- b) Give a detailed description of the most likely transformation that we applied to the original time series to obtain the plot of the transformed time series. (1)

Transformed time series:



- c) Investigate the ACF and PACF of the transformed time series. Based solely on your observations and knowledge of the special cases of the ARIMA model, what ARIMA model would you expect to perform the best in this case? Explain your reasoning in detail. (3)



- d) Based on the R output of the fitted models, identify the best fitted model (as fitted on the original time series) and explain which measure(s) you used to determine your answer. Do not only state the fitted model's letter (X, Y, or Z), but also state the actual $ARIMA(p, d, q)$ model fitted for your chosen model. (2)

MODEL X:

```
ARIMA(p,1,q)
Coefficients:
      ar1      ar2      ma1      ma2
      0.7421 -0.3564  0.4326 -0.5273
s.e.  0.1534  0.0669  0.1588  0.1574

sigma^2 = 1.277: log likelihood = -306.57
AIC=623.14  AICc=623.45  BIC=639.61
```

MODEL Y:

```
ARIMA(p,1,q)
Coefficients:
      ma1      ma2      ma3      ma4  intercept
      1.1490 -0.0746 -0.4205 -0.1793      0.1083
s.e.  0.0697  0.1090  0.1048  0.0690      0.1179

sigma^2 estimated as 1.263: log likelihood = -307.53, aic = 627.06
```

MODEL Z:

```
ARIMA(p,1,q)
Coefficients:
      ar1      ar2      ar3      ar4      ma1      ma2      ma3      ma4
      0.7317 -0.2089 -0.2067  0.0240  0.4367 -0.6753  0.0568  0.2058
s.e.  1.0563  1.2164  0.6917  0.2399  1.0542  0.5623  0.8473  0.3482
      intercept
      0.1058
s.e.  0.1230

sigma^2 estimated as 1.243: log likelihood = -305.89, aic = 631.79
```

SECTION C: REGRESSION

[20]

QUESTION 1

[6]

- a) A simple linear regression model is given by:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

In the above model, what is ϵ_i and why is it included in regression models? (2)

- b) What does the coefficient of determination measure in a simple linear regression analysis? (1)

- c) What assumptions are made about the residuals of a simple linear regression model that we must check when fitting these models? (3)

QUESTION 2

[8]

A production manager wants to create a model to predict the number of units produced by workers per shift. They collect the following data for 30 shifts:

- *units* – number of units produced during the shift.
- *time* – the time of day that the shift takes place in (either morning or afternoon).
- *workers* – the number of workers working during the shift.
- *temperature* – the ambient temperature during the shift (in degrees Celsius).

- a) Before fitting the model, the following correlation matrix was generated:

	units	workers	temperature
units	1.0000000	0.66687913	0.13102477
workers	0.6668791	1.00000000	0.05587483
temperature	0.1310248	0.05587483	1.00000000

Using this correlation matrix, which variable(s) do you suspect would be a good predictor of the number of units produced during a shift? Justify your answer. (2)

A model-building procedure is implemented to help choose which variables to include in the model. The results of this procedure are shown below:

```

Start: AIC=139.3
units ~ workers + temperature + time

              Df Sum of Sq   RSS   AIC
- temperature  1    39.14 2426.5 137.79
<none>                                2387.4 139.30
- time         1    290.92 2678.3 140.75
- workers      1   1742.69 4130.0 153.75

Step: AIC=137.79
units ~ workers + time

              Df Sum of Sq   RSS   AIC
<none>                                2426.5 137.79
- time         1    295.0 2721.5 139.23
- workers      1   1773.9 4200.4 152.25

Call:
lm(formula = units ~ workers + time)

Coefficients:
(Intercept)      workers  timemorning
      16.775         1.210        -6.468

```

- b) Are these results consistent with the correlation matrix in 2a? Explain why or why not by referencing all variables. (2)
- c) What is the AIC of the best model from the above model-building procedure? (1)
- d) The following results are obtained from fitting the model that incorporates all variables:

Coefficients:					
	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	10.3791	11.6181	0.893	0.379859	
workers	1.2008	0.2756	4.356	0.000184	***
temperature	0.2612	0.4001	0.653	0.519549	
timemorning	-6.4246	3.6094	-1.780	0.086772	.

Conduct a hypothesis test for the population parameter $\beta_{\text{temperature}}$ using a significance level of 5%. Clearly indicate your null and alternative hypotheses and justify your conclusion. (3)

QUESTION 3

[6]

A tech company wants to determine what makes users click on their app notifications. They collect the following data on 50 app users:

- *Click* – a binary variable that indicates whether a user clicked on the notification or not (“Yes”=1 or “No”=0)
- *Time* – time of day the notification is sent (“during work hours” = *TimeDuringWork* or “after work hours” = *TimeAfterWork*)
- *Last_interaction* – time since last app interaction (in hours)

A logistic regression model was fitted to this data, and the results are shown below:

```
Call:
glm(formula = Click ~ Time + Last_interaction, family = "binomial")

Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept)    0.31835    0.80090   0.397   0.6910
TimeAfterWork    1.39202    0.65778   2.116   0.0343 *
Last_interaction -0.02610    0.02284  -1.143   0.2531
---
```

- a) What is the interpretation of the *Last_interaction* coefficient in terms of odds? (2)
- b) Calculate the 95% confidence interval (on the log-odds scale) for the *Time* coefficient. (2)

- c) What types of variables are appropriate for a logistic regression model? Specifically, what form should the dependent variable take, and what forms can the independent variables take for the model to be suitable? (2)

SECTION D: ANOVA AND NON-PARAMETRIC TESTS

[37]

ANOVA

QUESTION 1

[5]

A team of food scientists is investigating how the method of cooking and the type of oil used affect the amount of fat retained in fried potato slices. They tested two cooking methods: air frying and deep-frying, and used three types of oils: sunflower oil, olive oil, and coconut oil. For each combination of cooking method and oil type, they prepared and tested two separate batches of potato slices. All samples were cooked under standardised conditions, and the amount of fat retained (in grams per 100g serving) was measured after cooking. Mean fat retained is shown in brackets.

Oil Type	Air Fryer			Deep Fryer		
Sunflower	7.2	6.9	(7.05)	11.4	11.8	(11.60)
Olive	6.3	6.5	(6.40)	5.7	5.4	(5.55)
Coconut	7.5	7.3	(7.40)	13.2	13.5	(13.35)

- a) Consider each combination of cooking method and draw an interaction plot with the available data: (3)
- Place oil type on the x-axis.
 - Draw separate lines for each cooking method and label each line clearly.
 - Label your axes clearly.
- b) Based on the shape of the plot, provide a brief comment on whether an interaction appears to be present. (1)
- c) Regardless of your answer to (b), explain what an interaction means in the context of this experiment. (1)

QUESTION 2

[5]

A product manager at SnapBasket, a grocery delivery startup, wants to determine whether the type of promotional discount offered to new users affects how much they spend on their first order. Three discount types were tested: a Flat Discount (R25 off the total basket), a Percentage Discount (10% off the basket total), and Free Delivery. The experiment was conducted using a completely randomised design. Fifteen new customers from the same delivery zone were randomly assigned to one of the three discount types, with five

customers per discount type. After the customers placed their first orders, the total amount spent (in Rand) was recorded, and the following R output was generated.

	Df	Sum Sq	Mean Sq	F value
DiscountType	2	6303	3151.7	66.54
Residuals	12	568	47.4	

- Write down the null and alternative hypotheses for this analysis. (1)
- Use an F-distribution table to estimate the p-value for the test. State the appropriate degrees of freedom and clearly explain your reasoning. (2)
- Based on your answer to the previous question, state your decision and conclusion in the context of the study, using a 5% significance level. (2)

QUESTION 3

[10]

ZingBuzz, a Cape Town-based startup that offers budgeting tools for young professionals, wants to boost app installations by promoting its new "Spending Insights" feature on Instagram. To evaluate which type of content is most effective, the marketing team tested three formats commonly used on Instagram:

- Feed Post (F): A static post with a caption highlighting the feature.
- Story (S): A vertical image with a swipe-up link and time-limited visibility.
- Reel (R): A short video showing how the feature works in real-time.

They suspect user engagement will vary by day of the week, especially in their target demographic, so the team ran the experiment over four weekdays (Monday to Thursday). Each day, the three formats were published simultaneously to separate audience segments using a specialised Instagram tool. The team measured engagement as the number of link clicks (i.e., taps on the install link) received within 24 hours.

Day	Feed Post	Story	Reel
Monday	38	51	64
Tuesday	42	54	67
Wednesday	37	49	61
Thursday	40	53	66

An analyst at the company conducted an ANOVA analysis and gave you the following information:

- The F-value for Day is 65.33.
- The F-value for Content Type is 2550.33.
- The residual sum of squares is 0.25.

- a) Identify the following based on the experiment description: (2)
- Treatment factor(s)
 - Treatments
 - Response
 - Number of replicates
- b) What is the treatment structure, and how has randomisation been conducted? (1)
- c) Based on the treatment structure and randomisation procedure you have identified, what type of design has been used here? (1)
- d) Write down the population model equation used for analysing data from this type of experiment and define each term. (1)
- e) What is the standard error of a treatment mean in this experiment? (1)
- f) What is the experimental unit in this study? Justify your answer briefly. (2)
- g) Was it necessary to conduct the experiment over multiple days? Justify your answer briefly. (2)

NON-PARAMETRIC TESTS

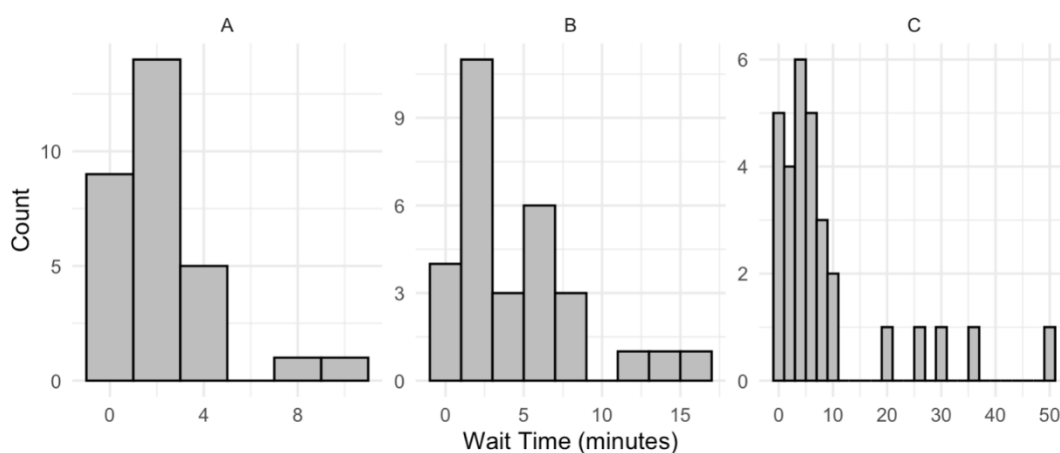
QUESTION 4

[8]

A logistics manager wants to compare customer wait times across three regional branches: Branch A, Branch B, and Branch C. She collects wait time data (in minutes) from 20 randomly selected customers at each branch during peak hours. A test was conducted. The test statistic was found to be 6.41 with a p-value of 0.0405.

Branch	Mean Wait Time (min)	Standard Deviation	Sample Size
A	10.4	2.1	30
B	14.3	3.9	30
C	11.6	11.5	30

Distribution of Wait Times by Branch



- a) Based on the summary statistics and plots provided above:
- Explain why a parametric test would not be appropriate in this case. (1)
 - Name the parametric test that would typically be considered for this scenario. (1)
- b) Name the appropriate non-parametric test for this situation and briefly justify your choice beyond the reasons given above. (2)
- c) Write down the null and alternative hypotheses for this analysis. (1)
- d) What is the distribution of the test statistic? (1)
- e) Interpret the result of the test at a 5% significance level using the p-value approach. (2)

QUESTION 5

[9]

A company wants to evaluate whether noise-cancelling headphones help employees focus better during deep work. Ten office workers each complete a 30-minute logic test, once with and once without noise-cancelling headphones. The test is scored out of 30 based on the number of correct responses. The test sessions are done on different days in randomised order to avoid fatigue or learning effects.

With Headphones	Without Headphones
25	22
21	19
24	24
20	18
22	21
26	23
16	17
23	20
21	19
24	22

- What type of data has been collected in this experiment, and what is the relationship between the samples? (1)
- What is the appropriate non-parametric test for this data based on your previous answer? (1)
- What possible reason might justify using a non-parametric test instead of a parametric test in this scenario? (1)

d) Perform the appropriate test using a 5% significance level. Marks will be awarded for:

- Stating the correct null and alternative hypotheses. (1)
- Writing down the correct steps for calculating the test statistic given below and obtaining its p-value. (3)

Test statistic = 58

Standardised test statistic = 2.58

(Significance level: 5%)

- Making the correct decision and conclusion. (2)

THE END